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# Homogeneous charge compression ignition(HCCI)

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# Outline of presentation:

- Need of HCCI
- HCCI engine concept
- Advantages and challenges related to HCCI
- HCCI prototypes
- Future aspects
- Conclusion

# Demerits of existing IC Engine

## SI Engine:

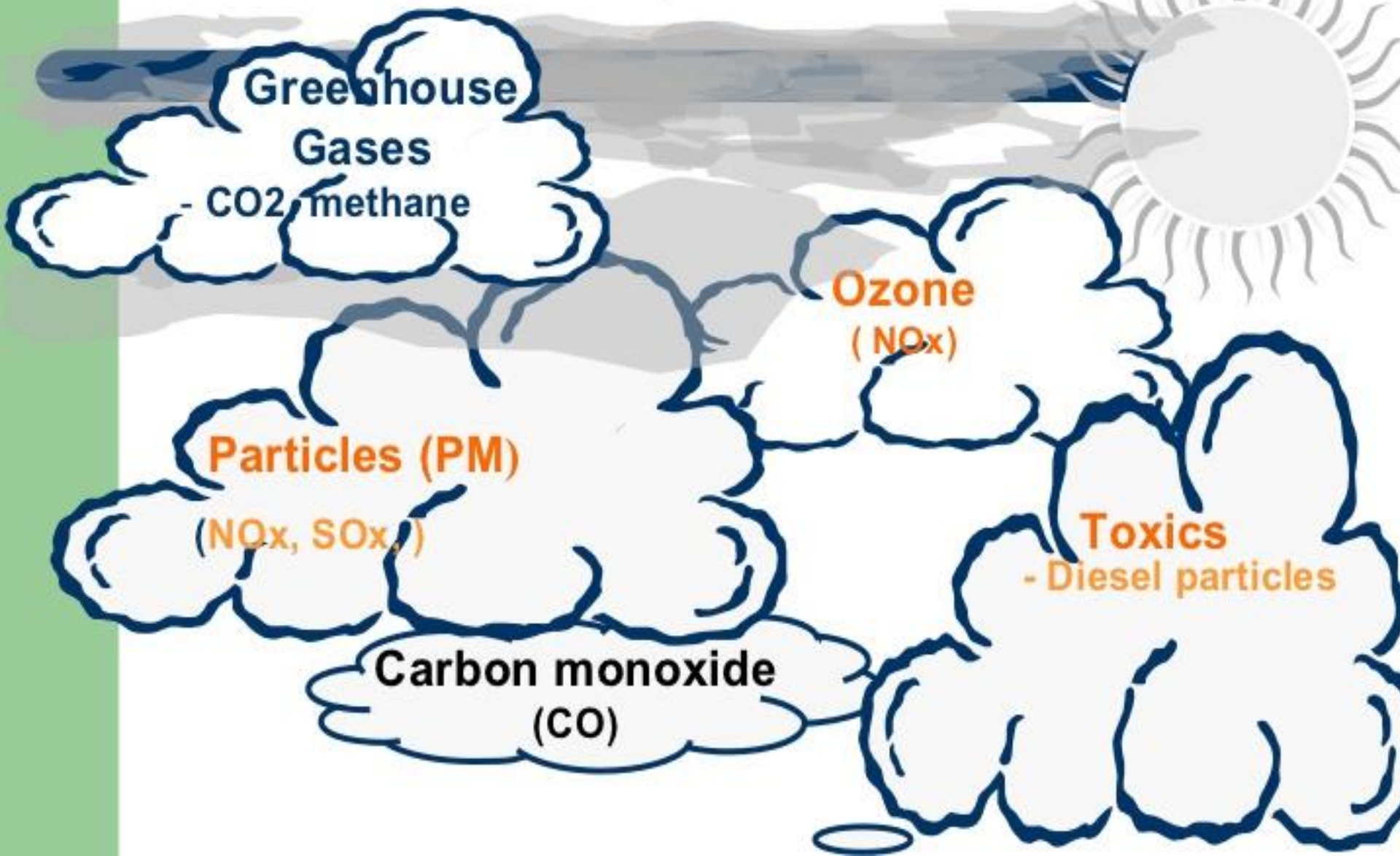
- Losses during gas exchanges
- Low thermal efficiency
- Low combustion efficiency

## CI Engine:

- Emission of Nox, soot particles and PM
- Diesel exhaust has been found to contain toxic air contaminants



# Diesel Engines Contribute to Multiple Air Pollutants





So  
homogeneous charge  
compression ignition engine

Is

The effective way to overcome  
the demerits.

# What is an HCCI Engine?

- HCCI is a form of internal combustion in which the fuel and air are compressed to the point of auto ignition.
- That means no spark is required to ignite the fuel/air mixture.
- Creates the same amount of power as other Engine, but uses less fuel.

# How Does It Work?

- A given concentration of fuel and air will spontaneously ignite when it reaches its auto-ignition temperature.
- Unlike conventional engines, the combustion occurs simultaneously throughout the volume rather than in a flame front.
- This important attribute of HCCI allows combustion to occur at much lower temperatures, dramatically reducing engine-out emissions of Nox.

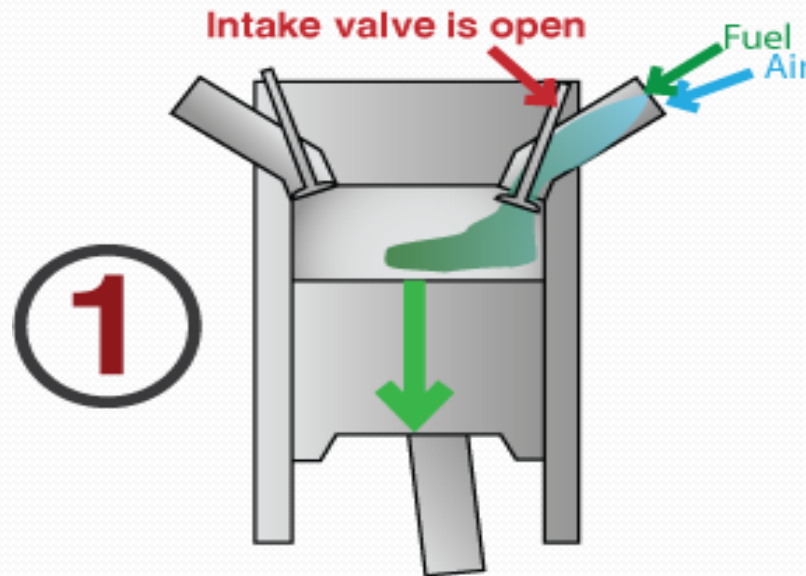


# Four Stroke Engine

- Suction Stroke
- Compression Stroke
- Combustion Stroke
- Exhaust Stroke

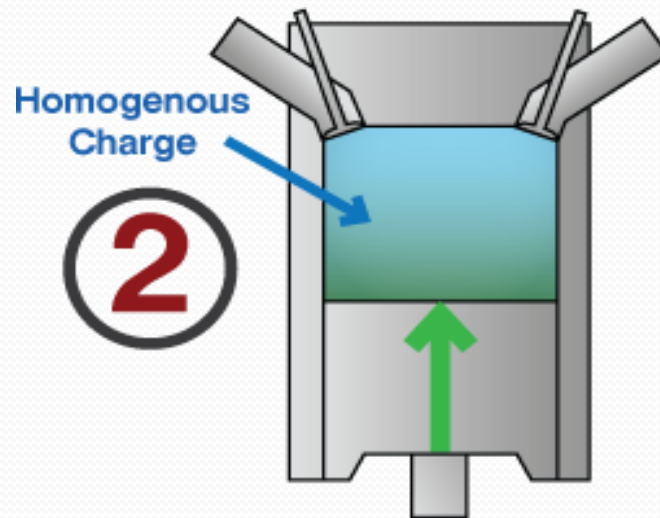
# Suction Stroke:

- Homogeneous charge or fuel-air mixture is drawn into the cylinder during this stroke.



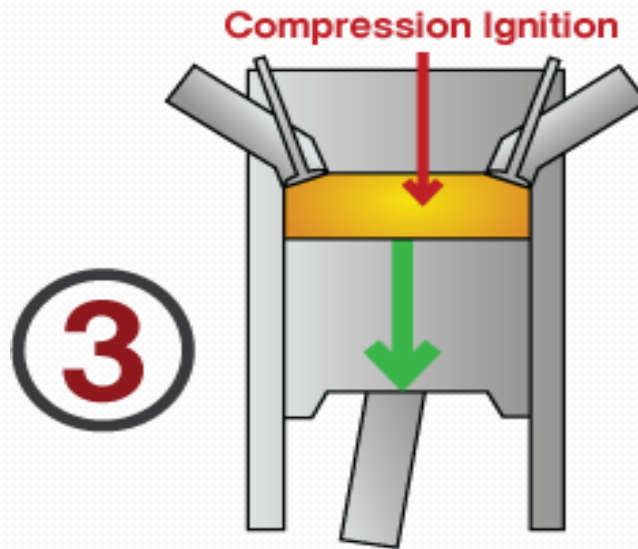
# Compression Stroke:

- Piston moves from bottom dead centre to top dead centre and homogeneous charge is compressed to high enough temperature and pressure during this stroke.



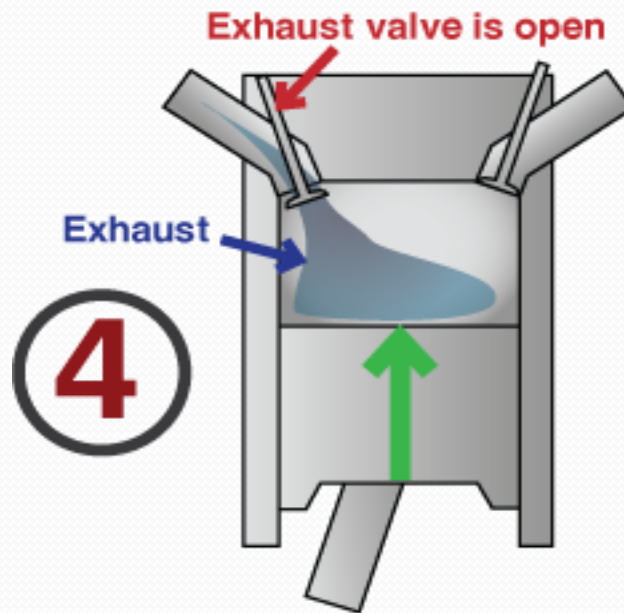
# Combustion Stroke:

- Combustion initiation occurs simultaneously at whole volume of combustion chamber and burn the fuel-air mixture.

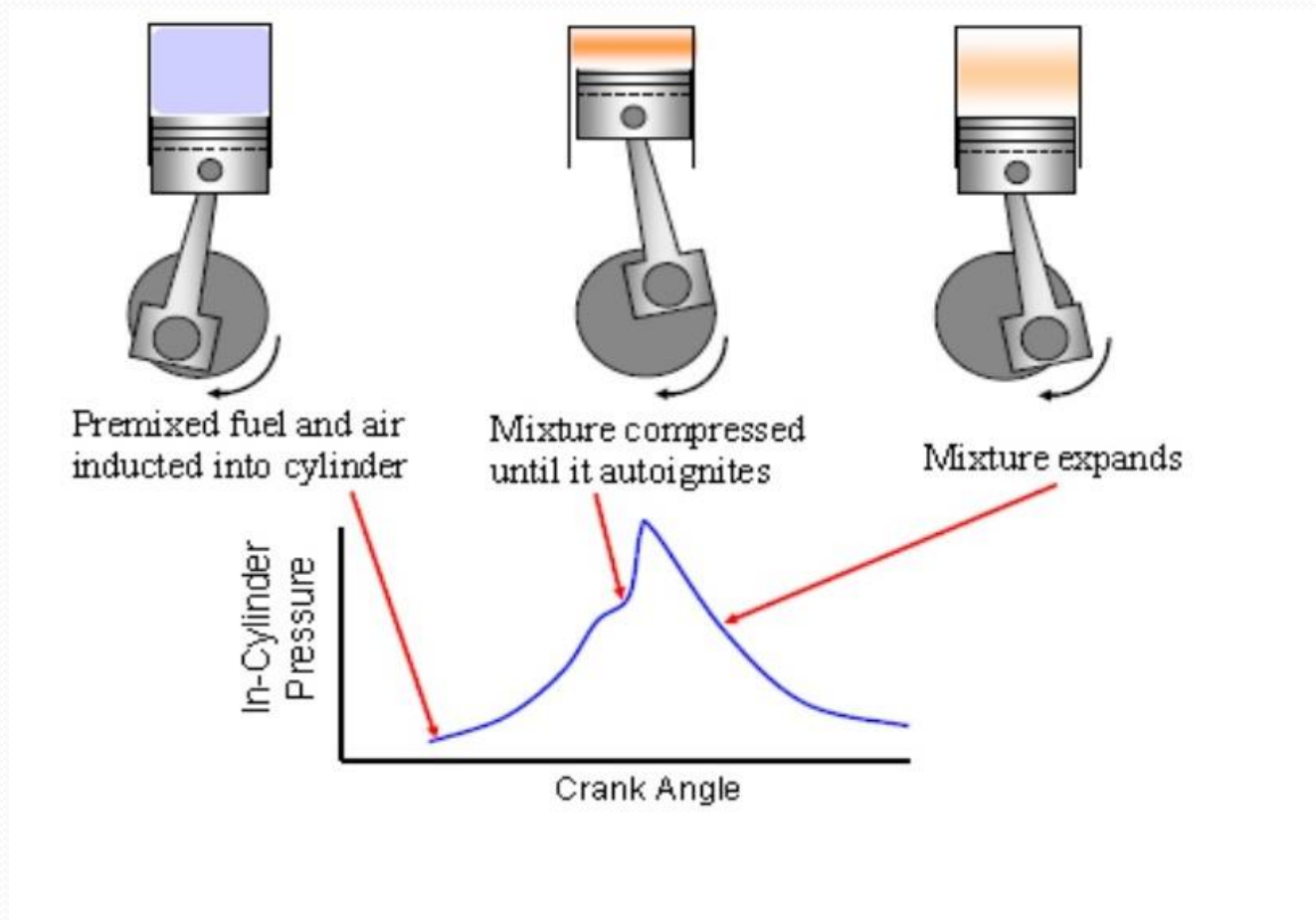


# Exhaust Stroke:

- Piston moves from bottom dead centre to top dead centre and removal of exhaust gas takes place through exhaust valve during this stroke.



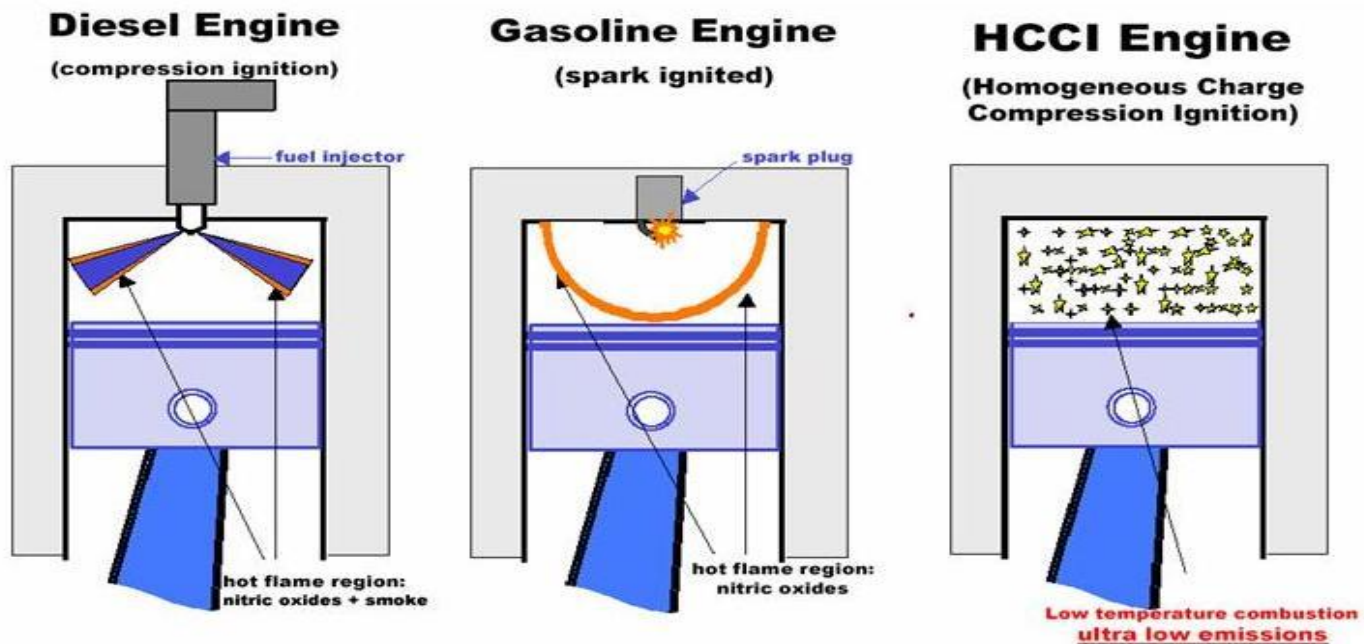
# Crank Angle Vs In-cylinder Pressure



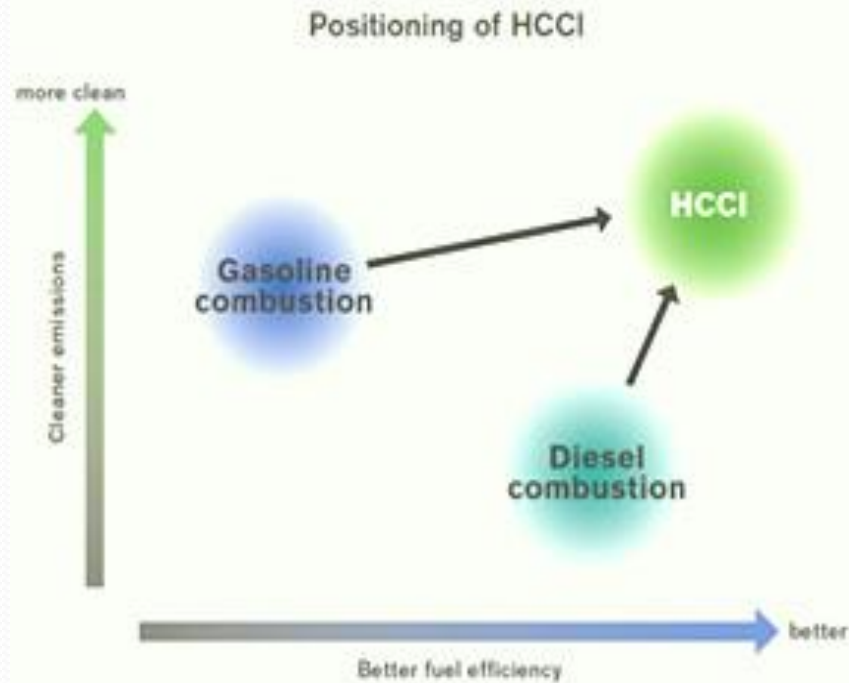


# Comparison With Other Engines:

- Unlike conventional engines, the combustion occurs simultaneously throughout the volume rather than in a flame front.



# Fuel Efficiency Vs Cleaner Emission



# Modes Of Operation:

- Combustion start is controlled externally in both SI engine and CI engine. while in HCCI, combustion start is dependent only on thermo-chemical conditions inside the cylinder.
- Practical HCCI engines will need to switch to a conventional SI or diesel mode at very low and high load conditions due to dilution limits.
- Two modes:
  - HCCI-DI dual mode
  - HCCI-SI dual mode

# HCCI – DI Dual Mode:

- A dual combustion system can potentially overcome the limits of low load operations and allow for gradual transition between conventional DI-mode at high load and HCCI mode at idle and low load.
- Combustion noise and Nox emissions were reduced substantially without an increase in PM.
- Thus DI-HCCI proves to be promising alternative for conventional HCCI with good range of operation.

# HCCI – SI Dual Mode:

- Generally, this dual mode engine combines the HCCI and SI together to achieve best performance.
- It equips VVA and spark ignition system
- Operates in HCCI mode at low to medium loads and switches into SI mode at higher loads.

# Control methods of HCCI combustion:

- The spontaneous combustion of fuel-air mixture need to be controlled.
- No direct control methods possible as in SI or CI engines.
- Various control methods are:
  - 1) Variable compression ratio
  - 2) Variable induction temperature
  - 3) Variable valve actuation

# Variable Compression Ratio Method:

- The geometric compression ratio can be changed with a movable plunger at the top of the cylinder head. This concept used in “diesel” model aircraft engine.
- Control of HCCI using variable compression ratio strategies has been shown effective results.

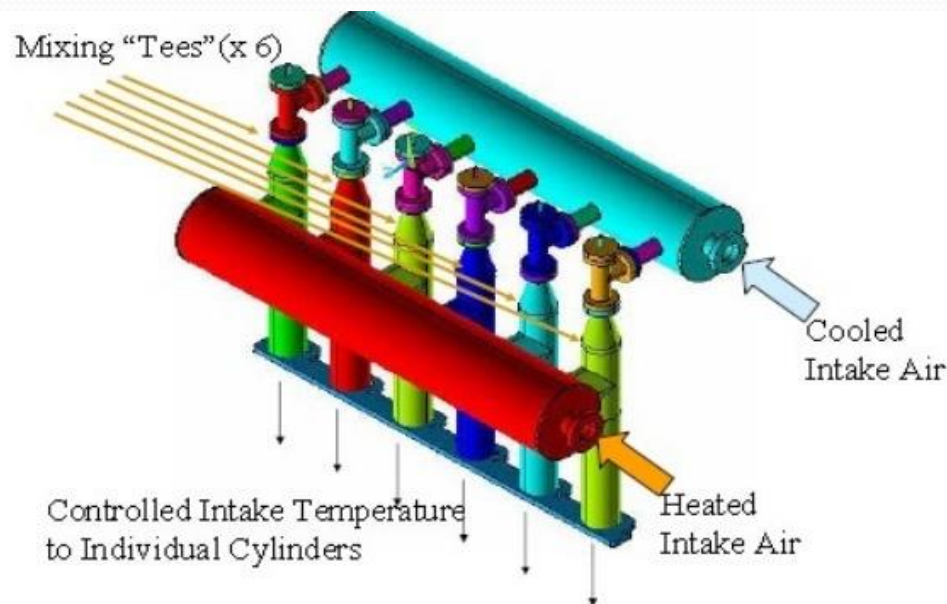


# Variable induction temperature method:

- The simplest method uses resistance heater to vary inlet temperature. But this method is slow.
- Now FTM (Fast Thermal Management) is used. It is accomplished by rapidly varying the cycle to cycle intake charge temperature by rapid mixing hot and cold air streams.

# Working Of FTM System:

- Rapid mixing of cool and hot intake air takes place achieving optimal temperature as demanded and hence better control.



# Variable Valve Actuation:

- This method gives finer control within the combustion chamber.
- Involves controlling the effective pressure ratio. It controls the point at which the intake valve closes. If the closure is after BDC, then the compression ratio change, altering the in-cylinder pressure time history prior to combustion.

# Advantages Of HCCI:

- 15% increase in fuel efficiency over SI engines.
- Can be operated at high compression ratios(>15).
- Can be operate on gasoline, diesel fuel and most alternative fuel.
- Have lower engine - out Nox and cleaner emission is possible.

# Disadvantages Or Challenges Of HCCI:

- High in-cylinder peak pressure.
- Engine wear due to high heat release and pressure rise.
- High level of noise.
- constrained at low loads and high loads.
- Cold start capability.

# Application of HCCI:

- GM used the HCCI combustion process in Opel Vectra. Which is powered by 2.2L ecotec engine that produces 55 mpg, i.e. 19.5 kpl.



# HCCI Prototypes:

- General Motors has developed a prototype called Saturn Aura with modified HCCI engines and upto 55 mph is achieved.





# HCCI PROTOTYPE:

- Mercedes-Benz has developed a prototype engine called Dies Otto, with controlled auto ignition. It was displayed in its F 700 concept car at the 2007 Frankfurt Auto Show. with a fuel consumption rating of 53 US MPG/22.35KPL.





# Future Of HCCI:

- The future of HCCI looks promising specially with partial HCCI mode.
- Major companies such as GM, Mercedes-Benz, Honda, and Volkswagen have invested in HCCI research.
- Mazda 3 is the upcoming HCCI based engine car and it will be released at the end of 2019.

# Conclusion:

- HCCI can be used as an alternative engine for CI and SI engines.
- 10-15% of thermal efficiency can be increased .
- HCCI engines are more efficient than SI and CI engines.
- It has fuel flexibility.
- Lower emission of PM and Nox to meet future norms.
- But HCCI Engines still needs a lot of development.



THANK YOU